Introduction & Hook

Ask:

"Why do some VR apps make users act instantly, while others fail to engage them?"

Explain:

VR is not just about graphics — it is about **human behavior and interaction**.

Animation Basics

Animation

Definition:

Animation is the process of creating motion by changing object properties over time.

Types:

- Keyframe animation
- Procedural animation
- Physics-based animation

Key Concept:

Interpolation creates smooth motion between frames.

Behavioral Modeling

Behavioral Modeling

Definition:

Behavioral modeling defines how objects or systems respond to user actions or environmental conditions.

Components:

- Input (user action/event)
- Logic (decision-making)
- Output (response)

Example:

User approaches object →
System detects →
Object reacts

Important Line:

Animation shows movement, behavior defines response.

Human Behavior Models in VR

Fogg Behavior Model (FBM)

Definition:

FBM states that behavior happens when three elements occur together:

$$Behavior = Motivation + Ability + Trigger$$

Components:

- **Motivation** → Why user wants to act
- **Ability** → How easy the action is
- **Trigger** → What prompts the action

VR Example:

In a VR fitness app:

- Motivation → Stay healthy
- Ability → Simple interface
- Trigger → Notification or prompt

If all 3 are present → user performs action.

Teaching Insight:

Good VR design ensures users are motivated, able, and triggered to act.

⏳ 45–52 min → Transtheoretical Model (TTM)

Definition:

TTM describes how users change behavior in stages.

Stages:

1. Precontemplation (not thinking about change)
2. Contemplation (thinking)
3. Preparation (planning)
4. Action (doing)
5. Maintenance (continuing)

VR Example:

VR rehabilitation system:

- User first observes exercises
- Then practices
- Then adopts routine

Key Idea:

VR systems can guide users through behavioral stages.

🔄 PART 4: Integration in VR Systems

Combining Everything

In VR:

- Behavior model → decides response
- Animation → shows action
- FBM → ensures user engagement
- TTM → supports long-term behavior change

Example:

VR learning app:

- Trigger → Quiz appears
- Behavior → Student answers
- Animation → UI responds
- TTM → improves learning habit

Recap & Questions

Key Points:

- Animation creates motion
- Behavioral modeling creates interaction
- FBM explains *why users act*
- TTM explains *how behavior evolves over time*

Quick Questions:

1. What is behavioral modeling?
2. What are the 3 components of FBM?
3. What is the purpose of TTM?
4. Difference between animation and behavior?